

Group no.8 - Team Members :

Engineering Ethics HUM 4441

Md. Mahiul Kabir
ID: 220041109

Rahatut Tahrim Mounota
ID: 220041122

Sohom Saboyob Sattyam
ID: 220041141

Imtiaz Morshed Adnan
ID: 220041148

Assignment Title

Ethical responsibilities of engineers and platforms in prioritizing user security, balancing cost and service - A Case Study on Telegram

Objective:

To critically examine the ethical responsibilities of engineers and digital communication platforms - focusing on Telegram - in balancing user security with cost-effectiveness and service delivery. Ethical frameworks will be applied to analyze real-world trade-offs and propose responsible practices.

Assignment Description:

1. Identify scenarios where user security may be compromised due to trade-offs involving platform features, cost limitations, or service design (e.g., Telegram's end-to-end encryption choices, metadata policies, and default privacy settings).
2. Research relevant case data using academic research, technical documentation, and journalistic reports about Telegram's privacy practices, past breaches, and developer choices.
3. Analyze these trade-offs through the lens of ethical theories (e.g., utilitarianism, deontology, virtue ethics) and existing ethical frameworks related to tech development (e.g., IEEE guidelines, ACM Code of Ethics).
4. Evaluate Telegram's decisions using ethical responsibility models, identifying who is accountable (e.g., developers, platform designers, policymakers).
5. Propose practical and ethical recommendations for platforms and engineers to better prioritize user security without severely compromising service or scalability.
6. Present key ethical dilemmas and recommendations visually using charts, ethical decision matrices, or flow diagrams.

Suggested Timeline

Week 1-2:

Identify where trade-offs affect security. Conduct literature review and contextual research (academic papers, tech blogs, news articles, Telegram documentation).

Week 3-4:

Analyze dilemmas using ethical theories and frameworks. Evaluate Telegram's key decisions and balance of stakeholder needs.

Week 5-6:

Finalize written report. Write key sections, develop relevant visuals and seek feedback.

Week 7-8:

Prepare a presentation highlighting key insights, challenges, and recommendations.

Suggested Research Sources:

- **Telegram FAQ and Policy Documents**
- **Academic journals** on cybersecurity ethics and platform governance
- **IEEE Code of Ethics** and **ACM Code of Ethics**
- **Center for Humane Technology** reports
- **News articles** on Telegram's privacy controversies (e.g., Wired, The Guardian, TechCrunch)
- **EU Guidelines** for platform accountability and secure communications